Parallel Programming: Assignment 3

Hybrid CPU-GPU Integer Sort

This assignment will get you started on the simultaneous use of both the CPU and GPU cores for computations.

The problem is of integer sort. Take a large array such that the size of the array is four times more than what can be accommodated in the GPU memory of the platform you will be using. Hence the array has to be divided between CPU and GPU. Use both the CPU and GPU executions simultaneously to sort their respective parts of the arrays, and finally merge sort them to produce the final sorted output for the entire array.

Compare the execution times of 1) CPU-only executions and 2) Hybrid CPU-GPU executions. Also, show CUDA profiling results.